AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Digital Realty, VA -- station KIAD, America/New_York
Committed pair	OSM 251716328 -> 251716311 Digital Realty / OSM way 251716311
Facade gap	408.0 m between the two halls
Weather record	43,763 real hourly records from KIAD
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.2784 C at 320 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2025-03-11 (crossing)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	1 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 2 of 24 hours with 1 mode change. The reactive incumbent took 13 hours -- 11 more -- but declared 1 unsafe hour against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 12 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	2 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	13 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s) the incumbent broke its own switch budget 1 time(s) to stay safe 12 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	4.174	18.0	3.934	1.096
01	mechanical	yes	switch budget	4.174	18.0	3.934	1.037
02	mechanical	yes	switch budget	3.894	18.0	2.824	1.206
03	mechanical	yes	switch budget	2.987	18.0	1.128	1.087
04	mechanical	yes	switch budget	2.789	18.0	0.604	1.035
05	mechanical	yes	switch budget	2.204	18.0	0.044	1.211

06	mechanical	yes	switch budget	0.844	18.0	-1.066	1.026
07	mechanical	yes	switch budget	2.509	18.0	0.044	1.679
08	mechanical	yes	switch budget	6.197	18.0	5.560	1.878
09	mechanical	yes	switch budget	9.911	18.0	11.336	2.401
10	mechanical	yes	switch budget	11.543	18.0	13.890	2.220
11	mechanical	yes	switch budget	15.394	18.0	17.780	1.718
12	mechanical	no	dry-bulb	18.206	18.0	19.440	1.324
13	mechanical	no	dry-bulb	19.799	18.0	21.110	1.067
14	mechanical	no	dry-bulb	22.174	18.0	21.933	1.336
15	mechanical	no	dry-bulb	22.566	18.0	22.780	1.008
16	mechanical	no	dry-bulb	22.956	18.0	22.780	1.129
17	mechanical	no	dry-bulb	22.610	18.0	22.220	1.333
18	mechanical	no	dry-bulb	22.076	18.0	21.110	1.268
19	mechanical	no	dry-bulb	20.680	18.0	18.890	1.344
20	mechanical	no	dry-bulb	19.916	18.0	17.780	1.558
21	mechanical	no	dry-bulb	18.462	18.0	15.560	1.654
22	FREE	yes	-	15.664	18.0	12.220	1.298
23	FREE	yes	-	15.637	18.0	13.330	1.035

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.174 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.174 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.894 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.987 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.789 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.204 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 0.844 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.509 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 6.197 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.911 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.543 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.394 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.206 C against a 18.0 C limit, so it fails by 0.206 C. It would take a limit 0.206 C higher, or a bound 0.206 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.799 C against a 18.0 C limit, so it fails by 1.799 C. It would take a limit 1.799 C higher, or a bound 1.799 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.174 C against a 18.0 C limit, so it fails by 4.174 C. It would take a limit 4.174 C higher, or a bound 4.174 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.566 C against a 18.0 C limit, so it fails by 4.566 C. It would take a limit 4.566 C higher, or a bound 4.566 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.956 C against a 18.0 C limit, so it fails by 4.956 C. It would take a limit 4.956 C higher, or a bound 4.956 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.610 C against a 18.0 C limit, so it fails by 4.610 C. It would take a limit 4.610 C higher, or a bound 4.610 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.076 C against a 18.0 C limit, so it fails by 4.076 C. It would take a limit 4.076 C higher, or a bound 4.076 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.680 C against a 18.0 C limit, so it fails by 2.680 C. It would take a limit 2.680 C higher, or a bound 2.680 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.916 C against a 18.0 C limit, so it fails by 1.916 C. It would take a limit 1.916 C higher, or a bound 1.916 C tighter, to change this hour.

21:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.462 C against a 18.0 C limit, so it fails by 0.462 C. It would take a limit 0.462 C higher, or a bound 0.462 C tighter, to change this hour.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.664 C, which is 2.336 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.298 C, and the margin is measured, not chosen: 1.189 C of group-conditional forecast error for this hour of day, plus 0.1086 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.637 C, which is 2.363 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.035 C, and the margin is measured, not chosen: 0.953 C of group-conditional forecast error for this hour of day, plus 0.0818 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

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